# Sahaj Bhadja

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## Summary

Software Engineer pursuing an MSCS at USC with 2+ years building distributed, low-latency, high-throughput systems and research infrastructure in C++, Java, and Python, with experience in data pipelines, simulation, and time-series financial data, and a strong interest in high-performance trading systems.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

## Experience

**USC Dornsife** **Los Angeles, CA**
*Software Engineer* *September 2025 – Present*

- Built **low-latency, high-throughput data pipelines** and distributed multi-tenant storage for 500+ users, optimizing queries and indexing to cut serving latency from 1.2s to 350ms.
- Engineered **distributed async orchestration** with explicit job-state transitions and failure isolation, processing 5K+ jobs and increasing throughput **3x** under concurrent load.

**Grad8** **Los Angeles, CA**
*Software Engineer Intern* *June 2025 – August 2025*

- Developed **high-throughput Python backend services** handling 10K+ daily requests, applying multithreading and concurrency for fault-tolerant, performance-critical batch processing.
- Collaborated in a **small, focused team** on code reviews and architecture, debugging performance bottlenecks to sustain stable latency under peak load.

**USC AD Labs** **Los Angeles, CA**
*ML Research Assistant* *January 2025 – May 2025*

- Built **research infrastructure and simulation pipelines** for offline reinforcement learning, processing 100K+ records and running closed-loop evaluations to support model development.
- Collaborated with **researchers** on a transformer-based model, improving evaluation accuracy **18%** and surfacing data patterns through large-scale experimentation.

**Aasma Technology Solutions** **India**
*Software Developer Engineer - (SecureSync)* *August 2023 – July 2024*

- Optimized **large-scale systems in Java and C++** with OOP design patterns, improving response times **30%** and building distributed ETL pipelines across three platforms.
- Automated **CI/CD, build, and 20+ test suites** (Jenkins, PyTest), performing root cause analysis and boosting pipeline efficiency **68%** in Agile sprints.

**LST Capitals** **India**
*Web Developer Intern* *November 2022 – January 2023*

- Built **time-series data pipelines** (Databricks, Spark) ingesting financial event data in micro-batches, delivering analytics and APIs at sub-200ms latency across 50K+ records.
- Integrated secure **financial transaction and webhook workflows** with idempotent handling, collaborating with stakeholders to translate requirements into reliable systems.

## Skills

**Languages**: C++, Java, Python, SQL, JavaScript/TypeScript, Bash
**Distributed & Performance**: Distributed Systems, Low-Latency, High-Throughput, Concurrency & Parallelism, Performance Optimization, Caching
**Data & Pipelines**: Data Pipelines, ETL, Time-Series Data, Real-Time Ingestion, Message Queues, Spark
**Research Infra & ML**: Simulation Pipelines, Reinforcement Learning, PyTorch, Backtesting-style Evaluation, Data Science
**Databases & Tools**: PostgreSQL, SQL Server, Elasticsearch, Kibana, MongoDB, Redis
**Cloud & Practices**: AWS, Kubernetes, Docker, CI/CD, Linux, Algorithms & Data Structures

## Projects

**CPU Scheduling Simulator** Live | *C++, React.js, OS Internals*

- Built a **C++ scheduling simulator** implementing six algorithms with performance analysis and real-time visualization, cutting average turnaround time **25%** across 1K+ simulated scenarios.

**FinNexus** Live | *Kubernetes, MongoDB, WebSockets, TypeScript*

- Built a **high-throughput financial platform** on Kubernetes processing 1M+ transactions for 50K+ users with real-time, low-latency data serving and token-based access.

**Big-Brain** Live | *GCP, PostgreSQL, FAISS, REST APIs*

- Built a **low-latency data pipeline** on GCP processing 50K+ records with semantic indexing and tuning that cut retrieval latency **40%**.